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//SIT 55 : ASSIGNMENT 9 : HEAP SORT
/*Implement Heap sort to sort given set of values using max or min
heap.*/
```

```
#include<iostream>
using namespace std;
#define SIZE 20
class List
{
public:
void heapSort(int a[], int n)
{
int i;
for(i=n/2; i>=1; i--)
{
adjust(a,i,n);
}
for(i=n; i>=2; i--)
{
int temp=a[i];
a[i]=a[1];
a[1]=temp;
adjust(a,1,i-1);
}
}
void adjust(int a[],int i, int n)
{
int j;
int no;
no=a[i];
j=2*i;
while(j<=n)
{
if(j<n && a[j] > a[j+1])
{
j=j+1;
}
if(no>a[j])
{
a[j/2]=a[j];
}
else
{
break;
}
j=j*2;
}
a[j/2]=no;
}
void displayMAX(int a[], int n)
{
cout<<"\nYour Max Heap: "<<endl;
for(int i=1;i<=n;i++)
{
cout<<a[i]<<" ";
}
}
void displayMIN(int a[], int n)
{
cout<<"\nYour Min Heap: "<<endl;
```

```

for(int i=n;i>0;i--)
{
cout<<a[i]<<" ";
}
};
int main()
{
List l;
int n;
int a[SIZE];
cout<<"Enter Size of list: ";
cin>>n;
cout<<"\nEnter array element : "<<endl;
for(int i=1;i<=n;i++)
{
cin>>a[i];
}
l.adjust(a,1,n);
l.heapSort(a,n);
l.displayMAX(a,n);
l.displayMIN(a,n);
}

```

/*OUTPUT:

Enter Size of list: 6

Enter array element :

92

55

4

21

8

96

Your Max Heap:

96 92 55 21 8 4

Your Min Heap:

4 8 21 55 92 96

*/